

PROBLEM STATEMENT

Question 1 – Problem Statement

A graph is given with nodes A, B, C, D, E, and G, where each edge has an associated path cost and each node has a heuristic value. The objective is to apply the A* Search Algorithm to find the optimal path from the start node A to the goal node G. During the dry run, each iteration should display the current node, Open List, Closed List, $g(n)$ (actual path cost), $h(n)$ (heuristic estimate), and $f(n) = g(n) + h(n)$ (evaluation function). After completing the search, identify the optimal path from A to G and calculate the total path cost. This problem evaluates the ability to trace the execution of the A* algorithm and verify its correctness using heuristic search.

Question 2 – Problem Statement

A game tree is provided with a MAX node at the root and two MIN nodes as its children, each containing leaf node values. The task is to perform a dry run of the Minimax Algorithm with Alpha–Beta Pruning by evaluating the tree from left to right. During the evaluation, record the Alpha (α) value, Beta (β) value, the selected value at each node, and identify any pruned nodes where further evaluation is unnecessary. Finally, determine the best move for the MAX player, state the final minimax value, and list all pruned nodes. This problem assesses the understanding of adversarial search and optimization using Alpha–Beta Pruning.